However, other factors may also apply. Higher mortality rates in high latitudes—where winters are harsh—may have led to the evolution of large clutch sizes. This is because the chances of survival until the next breeding season are low, and the reduced population results in more food being available for the survivors next season.

In 1982, Tore Slagsvold, a Norwegian evolutionary ecologist, advanced the nest predation hypothesis, which proposes that high rates of nest predation result in smaller clutches. If a nest with many chicks is found by a predator, more work by the parent birds will have been wasted than if the nest contained fewer chicks. Also, parents raising a large clutch are more likely to be seen by predators, because of the extra activity. Some ecologists have argued that the relative abundance of predators in the tropics has been more important than food supply in the evolution of small clutch sizes at low latitudes.

“Laying a clutch which will result in a smaller brood than … could be fed and reared successfully … confers advantages.”

Tore Slagsvold